



BODHI MULE HUNTER AI

Real-time mule account & suspicious transaction detection

Nine layers. Three complementary model views. One calibrated score an investigator can act on — and a containment layer that refuses to act without corroboration.

0.9966

FUSED ROC-AUC

0.9707

FUSED PR-AUC

99.99%

FALSE POSITIVES CUT

75x

PRECISION UPLIFT

0.054 ms

INLINE DECISION P50

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Team BODHI · CyberShield Hackathon 2026 - Problem Statement 2 · In association with Bank of India and IIT Hyderabad

THE PROBLEM

A mule account is a normal account

Real KYC. Real customer. Payments clear. Nothing about the account is anomalous — only the shape of the money moving through it, and the company it keeps.

HOW A MULE RING ACTUALLY WORKS

- 1 Victims** 40 defrauded people push money into one collector, in 90 minutes
- 2 Layering** A → B → C → D within minutes, each hop keeping 2–10%
- 3 Structuring** Split into transfers parked just under the reporting limit
- 4 Cash-out** Drained via ATM/AePS at 03:12 — out of the banking system

WHY TODAY'S RULE ENGINES FAIL

They score accounts in isolation, so they cannot express “this account's counterparties are themselves suspicious” or “these forty credits arrived in ninety minutes”. Lacking the vocabulary, they compensate with sensitivity.

9,647

ALERTS RAISED

1.33%

PRECISION

99 of every 100 investigations are wasted. That is the number we set out to fix.

Aggregates destroy the signal that matters

The same monthly totals describe a shopkeeper and a mule. Only the ordering separates them — and no aggregate preserves order.

SHOPKEEPER

“Received ₹4 lakh from 38 payers over the month.”

Spread across weeks. Money rests. Counterparties repeat.

MULE

“Received ₹4 lakh from 38 payers between 14:10 and 15:40, then withdrew ₹3.8 lakh in nineteen ATM visits beginning at 03:12.”

Identical aggregates. Completely different sequence.

So the ensemble is not three flavours of the same tabular model. It is a tabular model, a graph neural network and a temporal model — each seeing something the other two structurally cannot.

Nine layers, each owned by a named agent

Declared input/output contracts, so the provenance of every decision is recoverable from the audit trail.

INGEST & ENRICH		
L1	Transaction ingestion	UPI · IMPS · NEFT · RTGS · AePS · ATM · card + NCRP tickets + RBI feeds
L2	Feature engineering	60 features: velocity, fan-in/out, dormancy, burst, structuring, device reuse
L3	Graph construction	accounts × devices × IPs × VPAs, shared-identifier cliques capped
DETECT		
L4	XGBoost screening	tabular behaviour → exact TreeSHAP attributions
L5	GraphSAGE	network structure → rings, device farms, multi-hop layering
L6	Temporal graph network	event ordering → smurfing, dormant bursts, rapid routing
DECIDE & ACT		
L7	Risk fusion	monotone non-negative blend, isotonic calibrated 0-100
L8	Explainability	TreeSHAP + GNNExplainer + plain-English narrative
L9	Alerting & kill-switch	proportionate, corroborated, reversible, audited

Three views that cannot see each other's evidence

LAYER 4

XGBoost

SEES

Aggregates per account

CATCHES

Volume, ratios, cash-out share, structuring

BLIND TO

Who the counterparties are

AUC 0.9946

LAYER 5

GraphSAGE

SEES

Two-hop neighbourhood

CATCHES

Rings, device farms, multi-hop layering

BLIND TO

When things happened

AUC 0.9962

LAYER 6

Temporal GNN

SEES

Last 64 events, in order

CATCHES

Smurfing, dormant bursts, rapid routing

BLIND TO

Anything outside the account

AUC 0.9628

Both neural layers are implemented in **NumPy with hand-derived gradients** — no PyTorch, no CUDA, no 2 GB of wheels. Every analytic gradient is verified against central finite differences in the test suite, so “hand-rolled” does not mean “unverified”.

Fusion: two properties that are not negotiable

MONOTONICITY

Blend weights are constrained non-negative, so more evidence can never lower a score. An unconstrained stacker on a small fold reliably learns “higher temporal score means safer” — indefensible to an investigator, unshippable past a regulator.

CALIBRATION

The score drives an automated kill-switch at 85, so it has to mean what it says. Isotonic regression maps the blend onto observed frequencies.

0.00125

BRIER SCORE

0.0047

EXPECTED CALIB. ERROR

FOUR FOLDS, NOT THREE

train

fits the base models

val

early-stops them

fuse

fits the Layer-7 blend

test

touched exactly once

A stacker fitted on the fold that early-stopped its own inputs inherits their optimism — and came out worse than the best single layer.

RESULTS

Every layer, on held-out data

12,000 accounts · 834,738 transactions · 151 mules (1.26%) · 21 rings

LAYER	ROC-AUC	PR-AUC	P@1%	R@1%
XGBoost (L4)	0.9946	0.9540	0.983	0.781
GraphSAGE (L5)	0.9962	0.9481	0.983	0.781
TGN (L6)	0.9628	0.8036	0.892	0.709
Intelligence	0.6883	0.3774	0.475	0.378
Fused (L7)	0.9966	0.9707	0.992	0.788

The fused model beats every individual view on both ranking metrics — which is the complementarity claim, measured.

100%

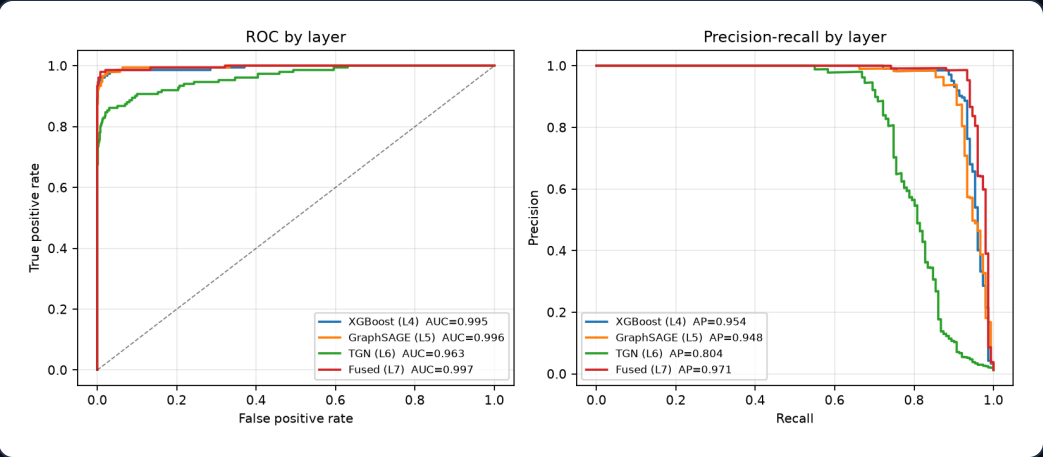
OUT-OF-TIME RECALL

64%

NEVER IN ANY NCRP
TICKET

0.113 ms

INLINE DECISION P99



Versus the rule engine it replaces

Compared at identical recall — because a system that alerts on everything can always claim to catch more.

RULE ENGINE · 8 SCENARIOS		→	BODHI · SAME RECALL	
Alerts raised	9,647		Alerts raised	129
True positives	128		True positives	128
False positives	9,519		False positives	1
Precision	1.33%		Precision	99.22%

99.99% of false positives eliminated at unchanged recall — a 75× precision uplift. The analyst clears the same true positives while opening 129 cases instead of 9,647.

We planted the accounts designed to break it

Any model scores ~1.0 AUC on naive synthetic fraud data. So the simulated bank is stocked with legitimate accounts that are structurally indistinguishable from mules.

LEGITIMATE POPULATION	COUNT	FALSE-POSITIVE RATE
Business Correspondent (Bank Mitra) agents	302	0.33%
Community collectors (chit fund, tuition)	107	2.80%
Small businesses with heavy fan-in	168	0.00%
Genuine dormancy reactivation	168	3.57%
Ordinary retail	10,504	0.77%

THE HARDEST DECOY

A Business Correspondent legitimately operates 8-22 village accounts from one handheld and dispenses AePS cash all day.

That is the exact fingerprint of a device farm doing rapid cash-out — and the system does not fall for it.

OTHER GUARDS AGAINST FLATTERING OURSELVES

Four disjoint folds
test touched exactly once

Cross-fitted seed model
no label leaks into a neighbour's average

Intelligence quarantined
NCRP tickets never reach the supervised layers

Point-in-time correctness
tested against a truncated ledger

WHERE IT IS WEAK

Recall is not uniform, and averaging hides that

A deployment team needs to know which typologies still need human-designed scenarios alongside the model.

TYPOLOGY / ROLE	RECALL	DETECTED
SMURFING	88.9%	16/18
role: CASHOUT	92.1%	35/38
FAN IN AGGREGATION	93.3%	14/15
LAYERING CHAIN	98.3%	58/59
role: COLLECTOR	98.3%	59/60
APK HARVEST	100.0%	15/15
DEVICE FARM	100.0%	31/31
DORMANT BURST	100.0%	13/13
role: RELAY	100.0%	53/53

WHY STRUCTURING IS THE HARDEST

A single sub-threshold transfer is, by construction, indistinguishable from a genuine near-threshold property or jewellery payment. Detection depends on an aggregate pattern that only becomes visible once enough transfers have accumulated.

AND CASH-OUT ACCOUNTS

Short, thin histories — they receive once and drain. There is simply less evidence to work with.

We report these rather than averaging them away. A prototype that hides its blind spots is not useful to the team that has to deploy it.

The investigator console

Four layer scores shown separately, evidence written in the vocabulary of an AML desk, cluster and exposure on every card.

B

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CRITICAL

143

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14

AT RISK

₹3,43,119

RESTRICTIONS

0

NCRP TICKETS

281

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BODHI Shield

Pipeline

Compliance

ALERT QUEUE

120

100

Fan In Aggregation - 5-account ring

ACC0004148 · ₹24,615 at risk

RING_MEMBERSHIP

TEMPORAL_PATTERN

OFF_HOURS

100

Fan In Aggregation - 9-account ring

ACC0008004 · ₹20,137 at risk

RING_MEMBERSHIP

HOURLY_VALUE

PAYER_RATE

1 NCRP

100

Fan In Aggregation - 5-account ring

ACC0003395 · ₹18,729 at risk

FIRST_TIME_PAYERS

RING_MEMBERSHIP

TEMPORAL_PATTERN

100

Fan In Aggregation - 5-account ring

ACC0009247 · ₹18,112 at risk

RING_MEMBERSHIP

TEMPORAL_PATTERN

OFF_HOURS

1 NCRP

100

Fan In Aggregation - 5-account ring

ACC0002587 · ₹17,410 at risk

RING_MEMBERSHIP

TEMPORAL_PATTERN

FIRST_TIME_PAYERS

CASE DETAIL

100

CRITICAL

ACC0004148

Fan In Aggregation - 5-account ring

Account ACC0004148 scores 100/100 (CRITICAL). One of 5 independently flagged accounts in a FAN_IN_AGGREGATION cluster that moved INR 533,390 between its members. The temporal model scores this account's event ordering at 99% - credits and debits are interleaved in a laundering pattern rather than a spending pattern. 8% of activity fell between midnight and 05:00. Cluster reference BODHI-RING-0046. Estimated exposure INR 24,615. Recommended: place a debit freeze pending customer contact and file an STR within 7 days if unexplained.

LAYER 4 · XGBOOST

99

tabular behaviour

LAYER 5 · GRAPHSAGE

100

network structure

LAYER 6 · TGN

99

event ordering

INTELLIGENCE

0

NCRP · APK · RBI

Evidence

Attribution

Network

Money trail

Actions

RING_MEMBERSHIP

Member of a detected fraud ring

One of 5 independently flagged accounts in a FAN_IN_AGGREGATION cluster that moved INR 533,390 between its members.

TEMPORAL_PATTERN

Abnormal event sequence

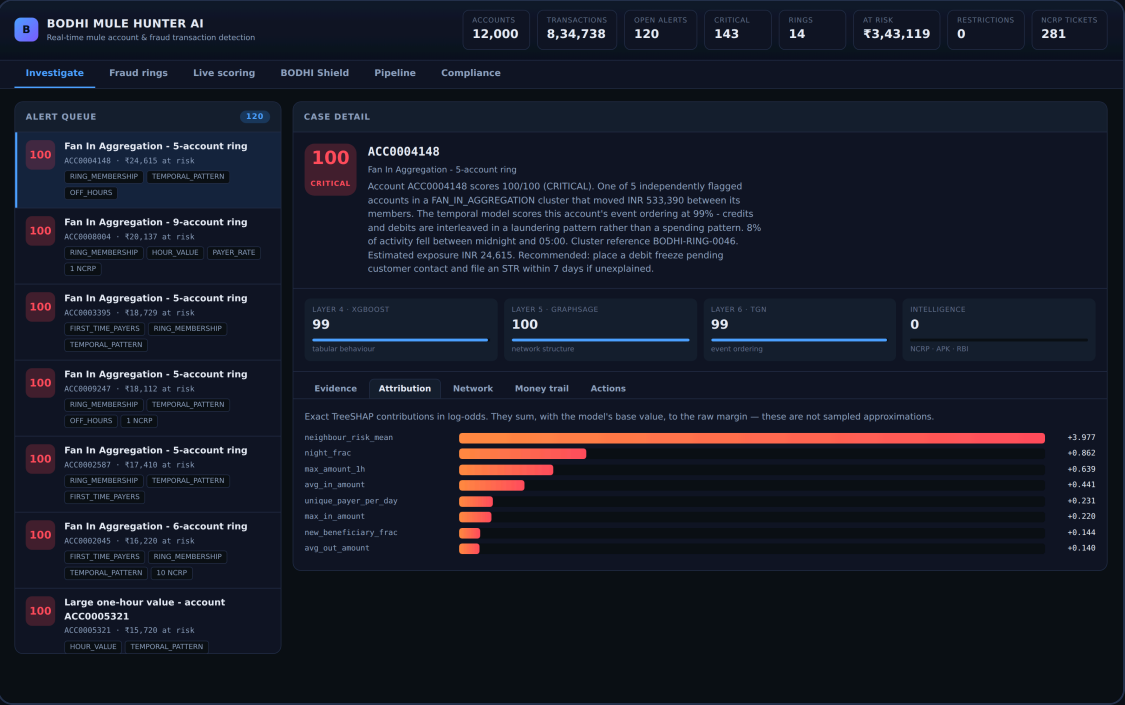
The temporal model scores this account's event ordering at 99% - credits and debits are interleaved in a laundering pattern rather than a spending pattern.

BODHI MULE HUNTER AI

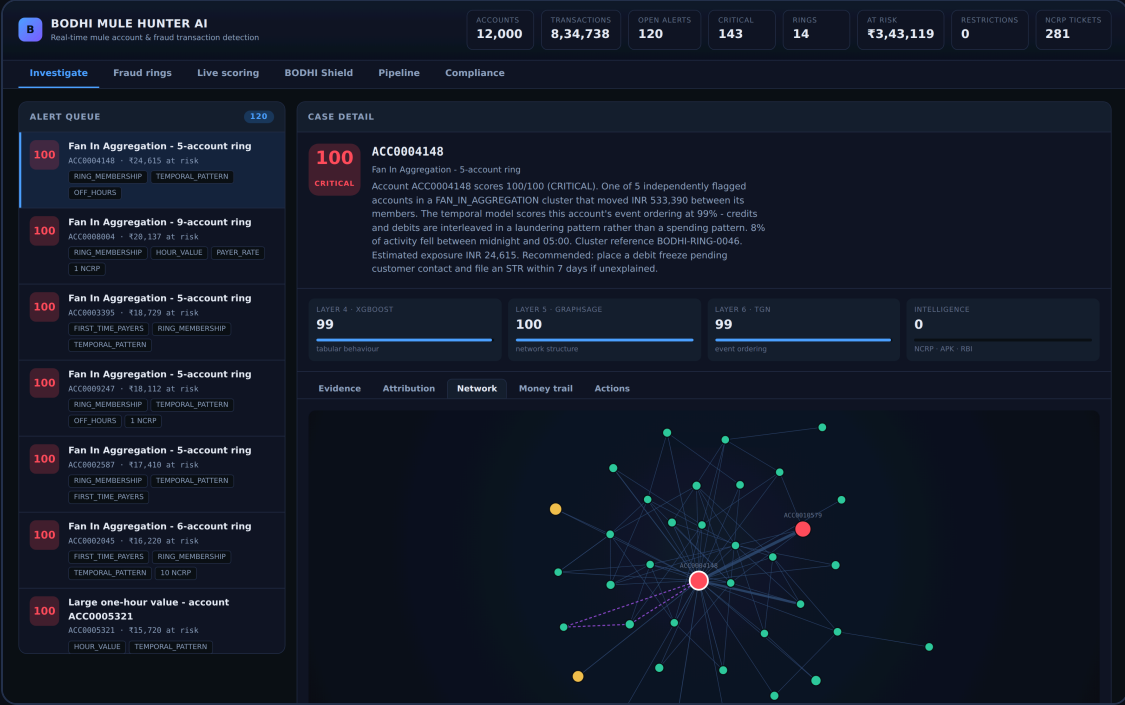
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Explanations specific enough to file



Exact TreeSHAP — attributions sum to the model's margin, not an approximation of it. Rendered as sentences, not feature names.



GNNExplainer answers which *relationships* drove the score. Dashed edges are shared-device links, invisible to any tabular model.

Time-respecting multi-hop trails

Every hop must occur after the previous one — money cannot be forwarded before it arrives. Value retention exposes each mule's commission.

B

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ACC0004148 · ₹24,615 at risk

RING_MEMBERSHIP · TEMPORAL_PATTERN

OFF_HOURS

100

Fan In Aggregation - 9-account ring

ACC0000004 · ₹20,137 at risk

RING_MEMBERSHIP · HOUR_VALUE · PAYER_RATE

1 NCRP

100

Fan In Aggregation - 5-account ring

ACC0003395 · ₹18,729 at risk

FIRST_TIME_PAYERS · RING_MEMBERSHIP

TEMPORAL_PATTERN

100

Fan In Aggregation - 5-account ring

ACC0009247 · ₹18,112 at risk

RING_MEMBERSHIP · TEMPORAL_PATTERN

OFF_HOURS · 1 NCRP

100

Fan In Aggregation - 5-account ring

ACC0002587 · ₹17,410 at risk

RING_MEMBERSHIP · TEMPORAL_PATTERN

FIRST_TIME_PAYERS

100

Fan In Aggregation - 6-account ring

ACC0002045 · ₹16,220 at risk

FIRST_TIME_PAYERS · RING_MEMBERSHIP

TEMPORAL_PATTERN · 10 NCRP

100

Large one-hour value - account

ACC0005321

ACC0005321 · ₹15,720 at risk

HOUR_VALUE · TEMPORAL_PATTERN

CASE DETAIL

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CRITICAL

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LAYER 4 · XGBOOST

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LAYER 5 · GRAPHSAGE

100

network structure

LAYER 6 · TGN

99

event ordering

INTELLIGENCE

0

NCRP · APK · RBI

Evidence

Attribution

Network

Money trail

Actions

Each hop must occur *after* the previous one and within 48 hours — money cannot be forwarded before it arrives. Value retention below 1.0 is the commission each mule takes.

ACC0004148

→

ACC0008629

→

ACC0001163

→

ACC0008398

3 hop(s) · ₹2,510 in, ₹123 out · 2932 min elapsed · median gap 1466 min · value retention 4.9%

ACC0004148

→

ACC0008629

→

ACC0001163

→

ACC0011527

3 hop(s) · ₹2,510 in, ₹1,411 out · 1560.3 min elapsed · median gap 780.2 min · value retention 56.2%

ACC0004148

→

ACC0008629

→

ACC0004896

→

ACC0001651

→

ACC0007961

4 hop(s) · ₹2,510 in, ₹221 out · 4980 min elapsed · median gap 1860 min · value retention 8.8%

ACC0004148

→

ACC0001671

1 hop(s) · ₹1,669 in, ₹1,669 out · 0 min elapsed · median gap 0 min · value retention 100.0%

ACC0004148

→

ACC0007976

1 hop(s) · ₹1,056 in, ₹1,056 out · 0 min elapsed · median gap 0 min · value retention 100.0%

THE TELL

A laundering chain leaks 2–10% at every hop as the handler's cut.

Legitimate payment chains simply do not have that shape — which is why value retention is shown on every path.

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Containment that can refuse

A wrongly frozen account is somebody unable to pay for medicine. No AUC improvement justifies being casual about that.

Proportionality

below 85 the automation cannot freeze at all

Corroboration

a full freeze needs ≥ 2 independent layers agreeing

Reversibility

every action carries a TTL and can be reverted

Rate limiting

bounded automated freezes per hour caps blast radius

Protected accounts

salary, pension and benefit accounts go to a human

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1

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120

100

Fan In Aggregation - 5-account ring

ACCB004148 - ₹24,615 at risk

RING_MEMBERSHIP | TEMPORAL_PATTERN

OFF_HOURS

100

Fan In Aggregation - 9-account ring

ACCB000044 - ₹28,137 at risk

RING_MEMBERSHIP | HOUR_VALUE | PAYER_RATE

1 NCRP

100

Fan In Aggregation - 5-account ring

ACCB003395 - ₹18,720 at risk

FIRST_TIME_PAYERS | RING_MEMBERSHIP

TEMPORAL_PATTERN

100

Fan In Aggregation - 5-account ring

ACCB009247 - ₹18,112 at risk

RING_MEMBERSHIP | TEMPORAL_PATTERN

OFF_HOURS | 1 NCRP

100

Fan In Aggregation - 5-account ring

ACCB002587 - ₹17,410 at risk

RING_MEMBERSHIP | TEMPORAL_PATTERN

FIRST_TIME_PAYERS

100

Fan In Aggregation - 6-account ring

ACCB002445 - ₹16,220 at risk

FIRST_TIME_PAYERS | RING_MEMBERSHIP

TEMPORAL_PATTERN | 10 NCRP

100

Large one-hour value - account

ACCB005321 - ₹15,720 at risk

HOUR_VALUE | TEMPORAL_PATTERN

100

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LAYER 4 - XGBBOOST

99

tabular behaviour

LAYER 5 - GRAPHUSAGE

100

network structure

LAYER 6 - TGN

99

event ordering

INTELLIGENCE

0

NCRP - API - RBI

Evidence

Attribution

Network

Money trail

Actions

Step-up auth

Hold credit

Freeze debit

Full freeze

Revert

Draft STR

Confirm fraud

Mark false positive

Recommended by the engine: FULL_FREEZE, REPORT_STR. The kill-switch will refuse or downgrade any action that lacks corroboration from at least two independent layers, and every refusal is audited.

```
{
  "account_id": "ACC0004148",
  "action": "FULL_FREEZE",
  "applied": true,
  "reason": "RISK 100/100 corroborated by 3 layers (xgb, graph, temporal)",
  "refusals": [],
  "expires_at": "2026-07-29T18:07:42.268999+00:00",
  "requires_human": false
}
```

When a constraint fires, the action is **downgraded** — never escalated — and flagged for human review. Every decision, including every refusal, is written to a hash-chained audit log whose head hash can be externally anchored.

BODHI SHIELD → BODHI MULE HUNTER

Malware analysis priming fraud controls before the first victim installs the app.

B

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APK STATIC TRIAGE

Upload an Android package. The manifest string pool is parsed, permission combinations are scored, and the DEX is mined for UPI handles, bank details and command-and-control endpoints. Everything financial it finds is pushed straight into the Mule Hunter graph.

Drop an .apk here, or

Choose File

 No file chosen

Analyse a bundled trojan sample

TRIAGE REPORT

100

CRITICAL

com.secure.bankassist

e4fd99b9b6cbe8e5e5e63634d68b24b0a751544ebc12ac733be566a4af517bd1

4 KB · 2 DEX · unsigned · obfuscation 83%

Toxic permission combinations

OTP interception combined with overlay phishing - the classic credential-theft pair

Enumerates banking apps and can drive their UI without the user

Can silently install and grant a second-stage payload

Findings

UNSIGNED

No v1 signature block; the package cannot be attributed to a publisher.

TOXIC_COMBINATION

OTP interception combined with overlay phishing - the classic credential-theft pair

TOXIC_COMBINATION

Enumerates banking apps and can drive their UI without the user

TOXIC_COMBINATION

Can silently install and grant a second-stage payload

PACKED

Commercial packer detected: Jiagu (360) packer.

OBFUSCATED

83% of identifiers are one or two characters - heavy renaming.

HARDCODED_VPA

3 UPI handle(s) hardcoded in the package - these are beneficiary accounts, streamed to BODHI MULE HUNTER AI.

HARDCODED_BANK_DETAILS

2 IFSC code(s) embedded.

HARDCODED_C2

2 bare IP address(es) - candidate command-and-control endpoints.

Financial indicators handed to Mule Hunter

UPI HANDLES

collect9821@ybl, payfast22@okaxis, recv0071@paytm

BANK ACCOUNTS

38291002847361, 50100234567891

IFSC CODES

HDFC0001234, SBIN0009876

C2 ADDRESSES

103.145.22.71, 45.88.190.14

1

Parse manifest

binary AndroidManifest.xml string pool, to spec

2

Score combinations

READ_SMS alone is a messaging app; + accessibility + overlay is a trojan

3

Mine the DEX

UPI handles, IFSC codes, accounts, C2 endpoints

4

Hand off

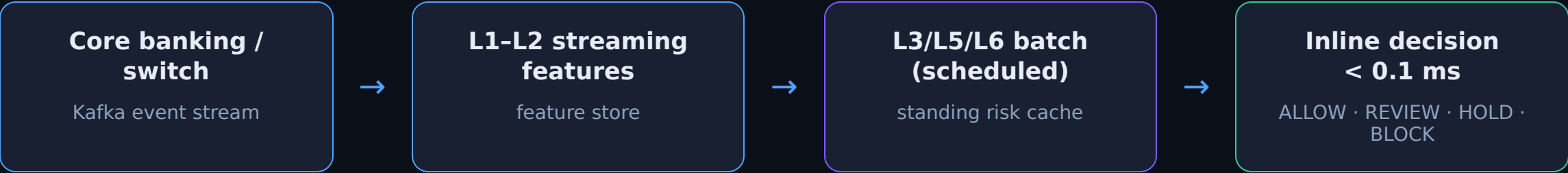
identifiers become weighted nodes in the fraud graph

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How it fits a real bank

The split between scheduled graph work and the inline path is what makes a UPI-latency decision possible at all.



MEASURED PERFORMANCE

Simulate the bank	7 s
Feature engineering	~4 s
Graph construction	~3 s
Full training, all layers	~123 s
Full population re-score	~38 s
Inline decision (p50)	0.054 ms

WHY THIS SPLIT

The graph and temporal layers run on a schedule and cache a standing risk per account. The authorisation path combines those cached scores with the transaction's own attributes and never traverses the graph — which is the only way a decision is feasible at payment latencies.

Built on the schema they published

3,924 declared columns, one row per alert. The names are machine-generated from a grammar, so we parse them instead of treating them as opaque.

FOUR COLUMNS LEAK THE LABEL

FRAUD_SUSPECTED · FALSE_POSITIVE · OTHER_RESOLUTION · UNATTENDED are resolution-status flags — how an analyst *closed* the alert. An open alert has none of them.

PR-AUC, quarantined (deployable) **0.177**

PR-AUC, resolution columns admitted **0.972**

FOUR STRATEGIES, SELECTION INSIDE EVERY FOLD

STRATEGY	FEATS	ROC-AUC	PR-AUC
Bank's 18 finalised	18	0.712	0.177
Bank + engineered	2,028	0.651	0.163
Automatic top- <i>k</i>	131	0.622	0.137
Every column	5,926	0.618	0.112

The bank's eighteen expert-chosen columns beat all 5,926. With 186 positives against thousands of predictors, that is what theory predicts.

0.7046
UNTOUCHED HOLDOUT ROC-AUC

0.7116
CROSS-VALIDATED ESTIMATE

5.5×
PR-AUC INFLATION IF LEAKED

3,924
COLUMNS PARSED BY GRAMMAR

Measured on a stand-in table with their exact schema — their data was not released when this was built. It proves the pipeline runs and does not flatter itself. It is not model performance, and we will not present it as such.

LIMITATIONS

Stated plainly, because hiding them helps nobody

Simulated data

Demonstrates the architecture beats a rule engine on data with realistic decoys. Not a forecast of production performance.

Static APK analysis only

The dynamic sandbox with runtime hooking needs an instrumented Android image; it cannot ship self-contained.

Graph layers are not real-time

~38 s full re-score. An account whose neighbourhood changed since the last batch is scored on slightly stale structure.

Single-institution view

Rings routing through several banks are only partly visible — a data-sharing problem, not a modelling one.

Cold start

The strongest feature is neighbourhood risk propagated from confirmed cases; with zero known mules performance drops materially.

Disparate impact unevaluated

Minimum-KYC and shared-device features are predictive but correlate with lower-income households. Alert-rate parity must be measured before go-live.

Mule detection fails today not because the signal is absent

— but because single-account rule engines cannot express it.

Three complementary views, fused under a monotonicity constraint into a calibrated score, cut false positives by 99.99% at unchanged recall — with explanations specific enough to file and containment cautious enough to automate.

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TESTS PASSING

~17.3k

LINES OF CODE

9

LAYERS, ALL RUNNING

0

EXTERNAL ML FRAMEWORKS

Akshay Tiwari

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Palak Vishwakarma

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Archi Singh Rajput

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Kartik Jain

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`make setup && make all && make serve` · every number on these slides is regenerated by that command